

Deep Watch

Waves & sound · systems engineering

A submarine — one line of ten compartments, hatches between them, and no sky at all.

45	15	5	6/10	3	16 · 28 · 1
STOPS	CHOICE (33%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

READING Where this campaign stands

The weakest curriculum delivery in the set after Aftershock and Wellmere: 4 of 10 equations computed, and none of the six gaps sits on an instrument, so Phase 0 will not rescue them. Snell's law is mentioned at five stops and computed at none — and concept 25, ray paths and shadow zones, is select-only. Two related facts, one fix. The place is the best in the set for this: a submarine is a single line of compartments, which is a chain you walk.

NOW The mix as it ships

CHOICE 15	BALLPARK 11	PROTOCOL 6	SEQUENCE 4	ROUTE 1	BALANCE 1	TRIAGE 1	DELEGATE 1	CONTROL 1	CHAIN 1	STRESS 1
ATTEST 1	ALLOCATE 1									

CURRICULUM What the syllabus is not getting

EQUATION THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
$v = f\lambda$	wave speed, frequency and wavelength	4 · PROTOCOL 6 · CHOICE 12 · CHOICE 15 · PROTOCOL 18 · CHOICE	verify first
$SL - 2TL + TS - NL \geq DT$	the sonar equation as a detection budget	36 · BALLPARK	real gap
$\sin\theta_1 / v_1 = \sin\theta_2 / v_2$	Snell's law — why sound bends at a layer	5 · CHOICE 9 · CHOICE 10 · SEQUENCE 18 · CHOICE 25 · CHOICE	real gap
$d = \frac{1}{2}vt$	echo ranging on a two-way travel time	8 · CHOICE 36 · BALLPARK	real gap

1 of 4 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 3 are unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM CONCEPT	STOPS
5	Resonance and harmonics	12
14	Thermodynamics and heat engines	32
25	Snell's law, ray paths and shadow zones	9

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects. The syllabus also lists 1 concept no stop touches at all — **Standing waves and cavity resonance** — which is expected, and is a different conversation from this one.

PROBE STRONG

stop 9 · SONAR

"Where the sound will not go" is a CHOICE. PROBE hands over no readings: take the sound-speed profile depth by depth and find where the ray bends away. The shadow zone is not the depth with the worst number on it, which is the format's own sentence.

SERVES Snell's law and concept 25 — the largest single gap in the game

NEEDS a profile with one honest break and a plausible decoy

SWEEP STRONG

stop 12 · ENG

"The one shaft speed that rings" is a CHOICE, and resonance is select-only. A resonance is the canonical SWEEP: drag shaft speed, watch vibration, and the peak is somewhere in a curve nobody drew for you.

SERVES concept 5, resonance and harmonics

NEEDS a narrow peak that is genuinely findable and not marked

HOLD STRONG

ENG or CTRL

Hold trim or depth inside a closing band while flooding pushes. This is the one thing the port from `deep_watch/` deliberately dropped — a rate that rises while you read a gauge had nowhere to live in walk-answer-hand-off. HOLD is where it lives.

SERVES buoyancy and control; recovers a simulation the port lost

NEEDS rate × time authored so a do-nothing run fails

TRIAL STRONG

the compartment line

Ten compartments, hatches between them, no sky. A casualty route driven in the order that does not spread the fire, graded on order and not the clock. The game already carries a ROUTE, which is the seated version of this.

SERVES casualty procedure and compartment dependency

RISK the boat is its own world module; gates need the theme's own geometry

SPOT STRONG

a SONAR watch stop

A standing order changed mid-watch, with nothing announcing it. Submarine watchkeeping runs on standing orders; the cost of missing a change is the format's whole subject.

SERVES watch discipline and contact classification

TALLY MEDIUM

stop 25 or 36

Pings accumulate into a bearing rate, and early on it moves because there is not enough behind it. Good, and it competes with the two SONAR BALLPARKs already there.

SERVES detection statistics

BELT REJECT

No line, no bench, no room for one.

LOB REJECT

Rejected on subject.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: `curriculumDelivery.mjs deepwatch --snapshot before.json`, then `--against before.json`.

Minimum honest 3 STOPS · ONE SITTING				
9	d3	SONAR	CHOICE → PROBE	Where the sound will not go the shadow zone, taken reading by reading
12	d4	ENG	CHOICE → SWEEP	The one shaft speed that rings the shaft speed that rings, found by looking
8	d3	NAV	CHOICE → BALLPARK	Two displays, one source echo ranging computed on a two-way time
CHOICE 15 → 12 (27%) · moves 16·28·1 → 13·29·3 · effective formats 5.0 → 5.8 · 2 new operate stops				

Balanced 6 STOPS · THE RECOMMENDATION				
9	d3	SONAR	CHOICE → PROBE	Where the sound will not go
12	d4	ENG	CHOICE → SWEEP	The one shaft speed that rings
8	d3	NAV	CHOICE → BALLPARK	Two displays, one source
32	d11	ENG	CHOICE → HOLD	One link is not handing the heat on the link that is not handing the heat on, held in band
25	d9	SONAR	CHOICE → TALLY	The rise on every bearing the rise on every bearing, accumulated before it is called
18	d6	ATMO	CHOICE → SPOT	Smoke is not only a breathing problem smoke, under an instruction that is replaced
CHOICE 15 → 9 (20%) · moves 16·28·1 → 10·29·6 · effective formats 5.0 → 7.0 · 5 new operate stops, 2 of them arcade				

Ambitious 9 STOPS · ADDS WORLD WORK				
9	d3	SONAR	CHOICE → PROBE	Where the sound will not go
12	d4	ENG	CHOICE → SWEEP	The one shaft speed that rings
8	d3	NAV	CHOICE → BALLPARK	Two displays, one source
32	d11	ENG	CHOICE → HOLD	One link is not handing the heat on
25	d9	SONAR	CHOICE → TALLY	The rise on every bearing
18	d6	ATMO	CHOICE → SPOT	Smoke is not only a breathing problem
10	d4	SONAR	SEQUENCE → TRIAL	What has to be true before the boat commits what has to be true before the boat commits — driven
5	d2	SONAR	CHOICE → CLOUD	The trace that will not resolve the trace that will not resolve, as a width against a limit
45	d15	SONAR	BALLPARK → SWEEP	Why the level pulses three times a second beats, as two frequencies you slide together
CHOICE 15 → 8 (18%) · moves 16·28·1 → 9·28·8 · effective formats 5.0 → 8.1 · 7 new operate stops, 3 of them arcade				

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument’s own board, and re-baseline curriculum-debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent.
2. Snapshot.
3. Edit the book, never the generated content — tools/import-book.mjs books/deep-watch.yml deepwatch --verify.
4. npm run traps: every instrument’s trap must fire.
5. npm run check deepwatch and npm run drive deepwatch, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark.
6. --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from `themes/deepwatch/content/` via `engine/dev/curriculumDelivery.mjs`; concept and equation lists authored in `tools/syllabus.js`. Format verdicts and slates are editorial. Run it with `THEME=deepwatch npm run dev`.